

Safety and efficacy of adjunctive cenobamate (YKP3089) in patients with uncontrolled focal seizures: a multicentre, double-blind, randomised, placebo-controlled, dose-response trial



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Summary

Background More than a third of patients with epilepsy are treatment resistant, and thus new, more effective therapies to achieve seizure freedom are needed. Cenobamate (YKP3089), an investigational antiepileptic drug, has shown broad-spectrum anticonvulsant activity in preclinical studies and seizure models. We aimed to evaluate the safety, efficacy, and tolerability of adjunctive cenobamate in patients with uncontrolled focal (partial)-onset epilepsy.

Methods We did a multicentre, double-blind, randomised, placebo-controlled, dose-response study at 107 epilepsy and neurology centres in 16 countries. Adult patients (aged 18–70 years) with focal seizures despite treatment with 1–3 antiepileptic drugs were randomly assigned (1:1:1:1) via an interactive web response system, by block sizes of 4 within each country, to adjuvant once daily oral cenobamate at dose groups of 100 mg, 200 mg, or 400 mg, or placebo following an 8-week baseline assessment. Patients, investigators, and study personnel were masked to treatment assignment. The study included a 6-week titration phase and 12-week maintenance phase. The primary efficacy outcomes were percentage change in 28-day focal seizure frequency (focal aware motor, focal impaired awareness, or focal to bilateral tonic-clonic seizures) from baseline analysed in the modified intention-to-treat population (≥ 1 dose and any post-baseline seizure data) and responder rates ($\geq 50\%$ reduction) analysed in the maintenance phase population (≥ 1 dose in the maintenance phase and any maintenance phase seizure data). The primary efficacy outcomes were analysed using a hierarchical step-down procedure comparing 200 mg versus placebo, 400 mg versus placebo, then 100 mg versus placebo. Safety and tolerability were compared descriptively across treatment groups for all randomised patients. This study is registered with ClinicalTrials.gov, number NCT01866111.

Findings Between July 31, 2013, and June 22, 2015, 437 patients were randomly assigned to either placebo ($n=108$) or cenobamate 100 mg ($n=108$), 200 mg ($n=110$), or 400 mg ($n=111$). Of these patients, 434 (106 [98%] in placebo group, 108 [100%] in 100 mg group, 109 [99%] in 200 mg group, and 111 [100%] in 400 mg group) were included in the modified intention-to-treat population, and 397 (102 [94%] in placebo group, 102 [94%] in 100 mg group, 98 [89%] in 200 mg group, and 95 [86%] in 400 mg group) were included in the modified intention-to-treat maintenance phase population. Median percentage changes in seizure frequency were -24.0% (IQR -45.0 to -7.0%) for the placebo group compared with -35.5% (-62.5 to -15.0% ; $p=0.0071$) for the 100 mg dose group, -55.0% (-73.0 to -23.0% ; $p<0.0001$) for the 200 mg dose group, and -55.0% (-85.0 to -28.0% ; $p<0.0001$) for the 400 mg dose group. Responder rates during the maintenance phase were 25% (26 of 102 patients) for the placebo group compared with 40% (41 of 102; odds ratio 1.97, 95% CI 1.08–3.56; $p=0.0365$) for the 100 mg dose group, 56% (55 of 98; 3.74, 2.06–6.80; $p<0.0001$) for the 200 mg dose group, and 64% (61 of 95; 5.24, 2.84–9.67; $p<0.0001$) for the 400 mg dose group. Treatment-emergent adverse events occurred in 76 (70%) of 108 patients in the placebo group, 70 (65%) of 108 in the 100 mg group, 84 (76%) of 110 in the 200 mg group, and 100 (90%) of 111 in the 400 mg group. Treatment-emergent adverse events led to discontinuation in five (5%) patients in the placebo group, 11 (10%) in the 100 mg dose group, 15 (14%) in the 200 mg dose group, and 22 (20%) in the 400 mg dose group. One serious case of drug reaction with eosinophilia and systemic symptoms occurred in the 200 mg cenobamate group. No deaths were reported.

Interpretation Adjunctive cenobamate reduced focal (partial)-onset seizure frequency, in a dose-related fashion. Treatment-emergent adverse events were most frequent in the highest dose group. Cenobamate appears to be an effective treatment option in patients with uncontrolled focal seizures.

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Research in context

Evidence before this study

We searched PubMed for randomised controlled trials published in English between Jan 1, 1994, and Jan 1, 2019, with the search terms “antiepileptic drug”, “partial epilepsy” OR “focal epilepsy”, “epilepsy”, “adjunctive”, and “clinical trial”. Efficacy results from pivotal placebo-controlled clinical studies of newer antiepileptic drugs for pharmaco-resistant focal epilepsy have shown significant, albeit incremental, improvement as add-on therapy. However, rates of seizure freedom, a clinically meaningful efficacy measure, have typically been low. Cenobamate is a newly developed antiepileptic drug under investigation for the treatment of focal epilepsy. One large, phase 2 clinical study has been done assessing adjunctive cenobamate (200 mg) in patients with uncontrolled focal seizures (NCT01397968). The results of the 12-week study are pending publication.

Added value of this study

This study is the first 18-week clinical study to assess the safety, efficacy, and tolerability of adjunctive cenobamate across a range of doses (100 mg/day, 200 mg/day, and 400 mg/day) compared with placebo in patients with uncontrolled focal seizures. Significant reductions in focal seizure frequency and significantly higher 50% or more

responder rates were observed in each dose group compared with the placebo group. When looking at the maintenance phase, these effects appeared to be dose-related. Notably, high seizure-free rates were shown during the maintenance phase in the 200 mg and 400 mg groups. Cenobamate was generally well tolerated, with most treatment-emergent adverse events rated as mild or moderate in severity. Treatment-emergent adverse events and withdrawals due to adverse events increased in a dose-related manner.

Implications of all the available evidence

This large, randomised, placebo-controlled, dose-ranging study provides safety, efficacy, and tolerability data for adjunctive cenobamate for patients with uncontrolled focal seizures. Both primary outcomes (median percentage seizure reduction and responder rate) were met, as required by the guidelines of the respective regulatory agencies. The current results also showed exceptional rates of seizure freedom with cenobamate in a clinical trial setting compared with other pivotal clinical studies of antiepileptic drug treatment in patients with uncontrolled focal seizures. These findings suggest that cenobamate might be a potentially useful treatment option for patients with uncontrolled focal epilepsy despite treatment with one to three antiepileptic drugs.

Introduction

Achieving seizure freedom is a primary goal in the treatment of epilepsy.¹ Because only half (50%) of treated patients with epilepsy become seizure-free with their first antiepileptic drug treatment,² many patients will require additional treatment to reach this goal. More than a third of patients with epilepsy, however, have uncontrolled seizures despite treatment with an antiepileptic drug,³ and few patients with drug-resistant epilepsy achieve seizure freedom with additional antiepileptic drugs, including newer drugs.²⁻⁴ Uncontrolled epilepsy is often disabling, with patients having increased comorbidity, psychological and social dysfunction, and an increased risk of premature death.^{5,6} Persisting generalised tonic-clonic seizures have been shown to be a risk factor for sudden unexpected death in epilepsy.⁷ Thus, there is an urgent need for new, more effective antiepileptic drugs to reduce seizures in patients with uncontrolled epilepsy and help more patients attain seizure freedom.

Cenobamate (YKP3089) is a novel tetrazole alkyl carbamate derivative⁸ that is in development and under investigation for the treatment of adults with uncontrolled focal (partial)-onset seizures. One of its mechanisms of action is thought to involve reduction of neuronal excitability by enhancing the fast and slow inactivation of sodium channels and by inhibiting the persistent component of the sodium channel current.⁹ Cenobamate has shown broad-spectrum anticonvulsant activity in preclinical studies of epilepsy and seizure models.⁸ A randomised, placebo-controlled, 12-week study (YKP3089C013; NCT01397968)

evaluated the safety and efficacy of cenobamate (200 mg) adjunctive treatment in patients with uncontrolled focal epilepsy despite treatment with one to three antiepileptic drugs. The results of the study are pending publication. In this study, we aimed to evaluate the safety, efficacy, and tolerability of cenobamate as adjunctive therapy across a range of doses in patients with uncontrolled focal epilepsy.

Methods

Study design and participants

This was a multinational, multicentre, double-blind, randomised, placebo-controlled, dose-response study, involving 107 epilepsy and general neurology centres from 16 countries (Australia, Bulgaria, Czech Republic, France, Germany, Hungary, Israel, Poland, Romania, Serbia, South Korea, Spain, Thailand, Ukraine, UK, and USA). This study was done in accordance with the International Council for Harmonisation Good Clinical Practice guidelines,¹⁰ as well as applicable government regulations and institutional research policies and procedures. The study protocol and amendments were reviewed and approved by an independent ethics committee or institutional review board in agreement with local requirements.

Eligible patients were adults aged 18–70 years with a diagnosis of focal epilepsy according to the International League Against Epilepsy's Classification of Epileptic Seizures.^{11,12} The epilepsy had to be uncontrolled despite treatment with at least one antiepileptic drug within the past 2 years. Patients must have been taking one to

three concomitant antiepileptic drugs at stable doses for at least 4 weeks before screening. To ensure consistency, all clinical data were reviewed and seizure diagnoses were confirmed by the Epilepsy Study Consortium before enrolment. Patients also had to have an EEG reading consistent with the diagnosis of focal epilepsy and CT or MRI scan done within the previous 10 years to rule out a progressive cause of epilepsy. Before randomisation, patients underwent a screening visit and an 8-week prospective baseline assessment to evaluate seizure frequency and type (recorded in patient diaries). During the 8-week baseline assessment, patients had to have eight or more focal aware (simple partial) seizures with a motor component, focal impaired awareness (complex partial) seizures, or focal to bilateral tonic-clonic (secondarily generalised) seizures, with a seizure-free interval of less than 25 days. Additionally, patients had to have at least three of these seizures during each of two consecutive 4-week segments of the 8-week baseline. Vagus nerve stimulation was permitted and did not count as an antiepileptic drug; however, the device must have been implanted at least 5 months before screening and the stimulation parameters must have been stable for at least 4 weeks before baseline.

Patients were excluded if they were taking diazepam, phenytoin, or phenobarbital within 1 month of screening because of the potential for drug–drug interaction with cenobamate. Patients were also excluded if they had taken vigabatrin within the past year, felbamate for less than 18 consecutive months, or intermittent rescue benzodiazepines more than once a month within the past month. Patients with a history of status epilepticus within 3 months of screening, clinically significant psychiatric illness or history of suicidal ideation within the past 6 months or suicide attempt in the past 2 years, or history of alcoholism or drug misuse within the past 2 years were also excluded. Additional exclusion criteria included a history of serious drug-induced hypersensitivity reaction or drug-related rash requiring treatment in hospital, antiepileptic drug-associated rash involving conjunctiva or mucosae, or more than one maculopapular rash requiring discontinuation. Full inclusion and exclusion criteria are provided in the appendix (pp 5–7).

We obtained written informed consent from each patient before study participation.

Randomisation and masking

We randomly assigned patients (1:1:1) to receive either cenobamate at 100 mg/day, 200 mg/day, or 400 mg/day, or a matching placebo. We did the randomisation centrally using an interactive web response system. Randomisation codes were computer generated by the designated statistician using SAS (version 9.2) with a block size of 4 within each study country to ensure equal allocation to each treatment group. The clinical team did not have access to the randomisation code. Patients, investigators, and study personnel were all masked to the treatment assignment.

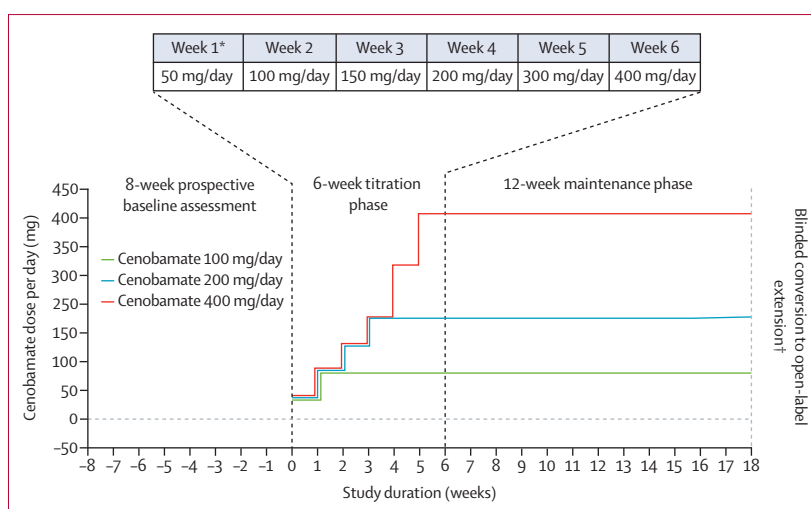


Figure 1: Amended study design

*The initial starting dose of cenobamate in the original faster titration schedule was 100 mg/day, with weekly increments of 100 mg/day to the target dose. The amended titration schedule reduced the initial starting dose to 50 mg/day and slowed the titration rate to the target dose to improve tolerability. †Target dose of 300 mg/day of cenobamate.

Study medications and packaging were identical in appearance to ensure adequate masking.

Procedures

Eligible patients were randomly assigned to the different groups and entered an 18-week double-blind treatment period (6-week titration phase and 12-week maintenance phase) followed by an optional open-label extension. During the titration phase, all patients began an initial starting dose of cenobamate at 100 mg/day orally that was up-titrated by a weekly increment of 100 mg to the target dose, or a matching oral placebo daily. Following a blinded review of the first nine patients, the study protocol was amended to lower the starting dose of cenobamate to 50 mg/day and slow the titration rate increase to 50 mg/week for tolerability, until reaching the target dose of 200 mg/day (figure 1). For patients randomly assigned to the 400 mg/day dose group, once the 200 mg/day dose was reached, the dose was up-titrated by 100 mg/day per week to the target dose. If a patient did not tolerate the next higher dose, the dose could be reduced to the previous dose. During the 12-week maintenance phase of the double-blind period, patients received their maximum attained dose of the study drug. Patients continued taking their allowed concomitant antiepileptic drug regimens unchanged throughout the entire double-blind treatment period. Following completion of the double-blind period, patients who continued to meet study eligibility, with the exception of the seizure frequency requirement, could enrol in an open-label extension.

See Online for appendix

Outcomes

On the basis of different regulatory requirements, the percentage change from baseline in focal seizure frequency

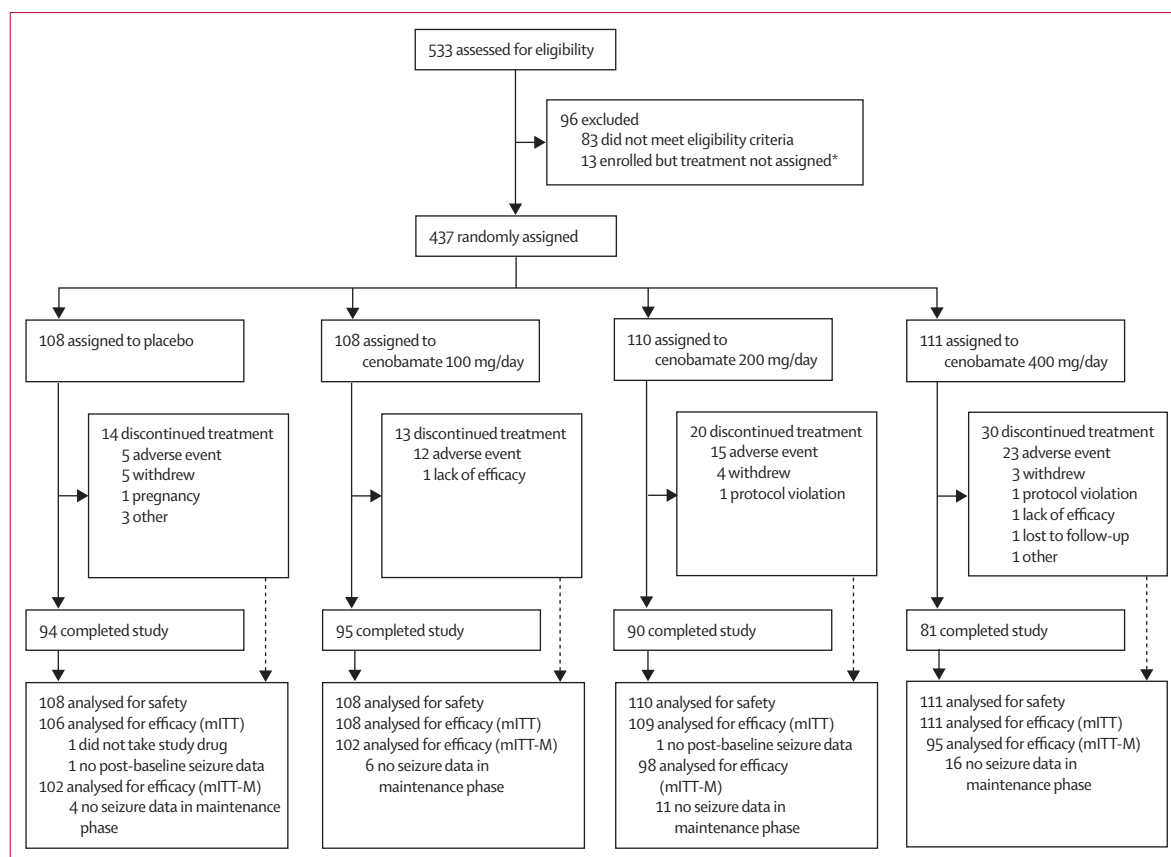


Figure 2: Trial profile

The intention-to-treat population and safety population included all patients randomly assigned to the treatment groups (n=437). The mITT population (n=434) included all patients randomly assigned to the treatment groups who had taken at least one dose of study drug and had any post-baseline seizure data. The mITT-M population (n=397) included patients who completed the 6-week titration phase and took at least one dose of study drug in the 12-week maintenance phase and had maintenance phase seizure data. mITT=modified intention-to-treat. mITT-M=modified intention-to-treat maintenance phase. *Eligible patients withdrawn from the trial before being assigned to a treatment group.

(focal aware motor, focal impaired awareness, or focal to bilateral tonic-clonic seizures) averaged over 28 days in the 18-week double-blind treatment period was considered the primary efficacy outcome for the US Food and Drug Administration, and the responder rate (ie, percentage of patients achieving $\geq 50\%$ reduction from baseline in focal seizure frequency) during the 12-week maintenance phase of the double-blind treatment period was considered the primary efficacy outcome for the European Medicines Agency.

Secondary efficacy outcomes included the percentage change in seizure frequency during the 12-week maintenance phase and the responder rate for patients during the 18-week double-blind treatment period. Additional responder rates, including 75% or more, 90% or more, and 100% reduction in seizures and percentage change from baseline in seizure frequency per 28 days by seizure type (focal aware with motor component, focal with impaired awareness, or focal to bilateral tonic-clonic seizures), were also assessed as prespecified outcomes. Cenobamate plasma concentrations were obtained under steady-state conditions at weeks 10 and 14 during the

12-week maintenance phase. Safety outcomes included the incidence of treatment-emergent adverse events, serious adverse events, discontinuations due to adverse events, and clinical laboratory evaluations, as well as changes from baseline in vital signs, 12-lead electrocardiograms (ECGs), physical and neurological examinations, and the Columbia-Suicide Severity Rating Scale responses.

Statistical analysis

Sample size estimation was based on data (publication pending) from a clinical study of adjunctive cenobamate (200 mg/day) in patients with uncontrolled focal seizures (YKP3089C013; NCT01397968). Assuming a standard deviation of 40%, as in YKP3089C013, a sample size of 400 patients (100 per treatment group) was estimated to provide 80% power to detect a 16% difference (average of differences seen in other focal seizure studies^{13–16}) in the seizure frequency percentage reduction between the cenobamate groups and placebo group at a two-sided significance level of 0.05 using an independent samples *t* test. A separate power calculation was not performed for the EMA outcome (responder rate). Potential loss to

follow up was not included in the sample size calculation. The modified intention-to-treat population included all randomly assigned patients who had taken at least one dose of study drug and had any post-baseline seizure data. The modified intention-to-treat population in the 12-week maintenance phase included patients who completed the 6-week titration phase and took at least one dose of study drug in the maintenance phase and had maintenance phase seizure data. The safety population included all randomly assigned patients.

Available seizure data were converted to a 28-day rate by summing the number of seizures in each period (eg, baseline, double-blind, and maintenance periods), dividing by the total duration exposed (days), and multiplying by 28. Because only non-missing seizure data were included, there was no specific adjustment for dropouts. The primary efficacy outcome of percentage change in seizure frequency per 28 days was assessed using an ANCOVA model fit to the ranked values of baseline seizure rate and treatment group. For the primary efficacy outcome of responder rate ($\geq 50\%$ reduction in seizures), data were analysed using a Fisher's exact test. A step-down procedure was used for the primary efficacy analyses to ensure the type I error rate for multiple comparisons was controlled at the 5% level using the following hierarchy: 200 mg versus placebo, 400 mg versus placebo, 100 mg versus placebo, in which a statistically significant difference had to be detected in this order for each subsequent comparison to occur.

Additional responder rates ($\geq 75\%$, $\geq 90\%$, and 100% seizure reduction) in the double-blind period and maintenance phase were analysed using Fisher's exact tests for treatment comparisons versus placebo. A prespecified subgroup analysis of percentage changes for focal seizure subtypes was analysed descriptively. Cenobamate plasma concentrations and safety and tolerability were summarised descriptively. Adverse events were coded according to the Medical Dictionary for Regulatory Activities (version 18.1).

Treatment comparisons versus placebo by seizure subtypes were analysed post hoc using an ANCOVA model with terms for ranked baseline seizure rate and treatment group. A post-hoc analysis of responder rates by focal seizure subtypes was also done and data were summarised descriptively. Seizure frequency and seizure-free rates over time during the 18-week double-blind treatment phase were analysed post hoc using sequential 4-week intervals (weeks 1–4, weeks 5–8, weeks 9–12, weeks 13–16, and weeks 15–18), with comparisons for each treatment group versus placebo using an ANCOVA model and Fisher's exact test, respectively, as previously described.

This study was overseen by an independent data monitoring committee, which did the unblinded safety reviews of the following: first 40 patients completing titration, first 80 patients completing titration, first 120 patients completing double-blind treatment, and first 240 patients completing double-blind treatment. We used SAS (version

	Placebo (n=108)	Cenobamate		
		100 mg (n=108)	200 mg (n=110)	400 mg (n=111)
Age (years)	39.6 (12.4)	39.0 (12.1)	40.9 (12.4)	39.6 (10.3)
Sex				
Men	58 (54%)	57 (53%)	54 (49%)	52 (47%)
Women	50 (46%)	51 (47%)	56 (51%)	59 (53%)
Race				
White	93 (86%)	89 (82%)	94 (85%)	96 (86%)
Asian	9 (8%)	10 (9%)	11 (10%)	11 (10%)
Black or African American	4 (4%)	4 (4%)	3 (3%)	1 (<1%)
Other	2 (2%)	5 (5%)	2 (2%)	3 (3%)
Body-mass index (kg/m ²)	27.4 (7.9)	26.0 (5.4)	26.1 (5.4)	25.8 (4.9)
Time since diagnosis (years)	23.0 (14.2)	25.5 (13.4)	22.8 (13.2)	24.4 (14.2)
Seizure type by history*				
Focal aware non-motor	24 (22%)	23 (21%)	20 (18%)	24 (22%)
Focal aware motor	22 (20%)	25 (23%)	25 (23%)	22 (20%)
Focal impaired awareness	84 (78%)	89 (82%)	84 (76%)	88 (79%)
Focal to bilateral tonic-clonic	60 (56%)	69 (64%)	61 (55%)	72 (65%)
Baseline seizure frequency per 28 days†	106	108	109	111
Median (IQR)	8.4 (6.0–19.0)	9.5 (6.0–19.8)	11.0 (6.0–26.0)	9.0 (6.0–21.5)
Number of previous antiepileptic drugs‡	3 (2–4)	3 (2–4)	3 (2–4)	3 (2–4)
Number of concomitant antiepileptic drugs§				
1	27 (25%)	25 (23%)	39 (36%)	24 (22%)
2	54 (50%)	48 (44%)	47 (43%)	62 (56%)
3	27 (25%)	34 (31%)	24 (22%)	24 (22%)
>3	0	1 (<1%)¶	0	1 (<1%)¶
Concomitant antiepileptic drugs				
Levetiracetam	41 (38%)	47 (44%)	48 (44%)	50 (45%)
Lamotrigine	31 (28%)	44 (41%)	27 (25%)	36 (32%)
Valproate or valproic acid	31 (28%)	23 (21%)	28 (26%)	28 (25%)
Carbamazepine	39 (36%)	29 (27%)	29 (25%)	25 (23%)
Oxcarbazepine	13 (12%)	15 (14%)	17 (16%)	19 (17%)
Clobazam	5 (5%)	17 (16%)	12 (11%)	17 (15%)

Data are mean (SD), n (%), or median (IQR), unless otherwise specified. *Patients might be reported in more than one category. †Calculated by the number of seizures over the baseline period divided by number of days in the interval multiplied by 28 (modified intention-to-treat population). ‡Antiepileptic drug medications taken any time before the start of the study; these might or might not have been ongoing during the study. §Antiepileptic drug medications ongoing at the start of the study and continued during the study. ¶Patient received temporary treatment with a fourth antiepileptic drug. ||Antiepileptic drug used in 10% or more of all patients.

Table 1: Baseline demographic and clinical characteristics of randomly assigned population

9.2) for all statistical analyses. This study is registered with ClinicalTrials.gov, number NCT01866111.

Role of the funding source

The funder of the study was responsible for the study design and statistical analysis, and MK, an employee of the sponsor, as an author had a role in data analysis, data interpretation, and writing of the report. The corresponding author had full access to all the data in the study and

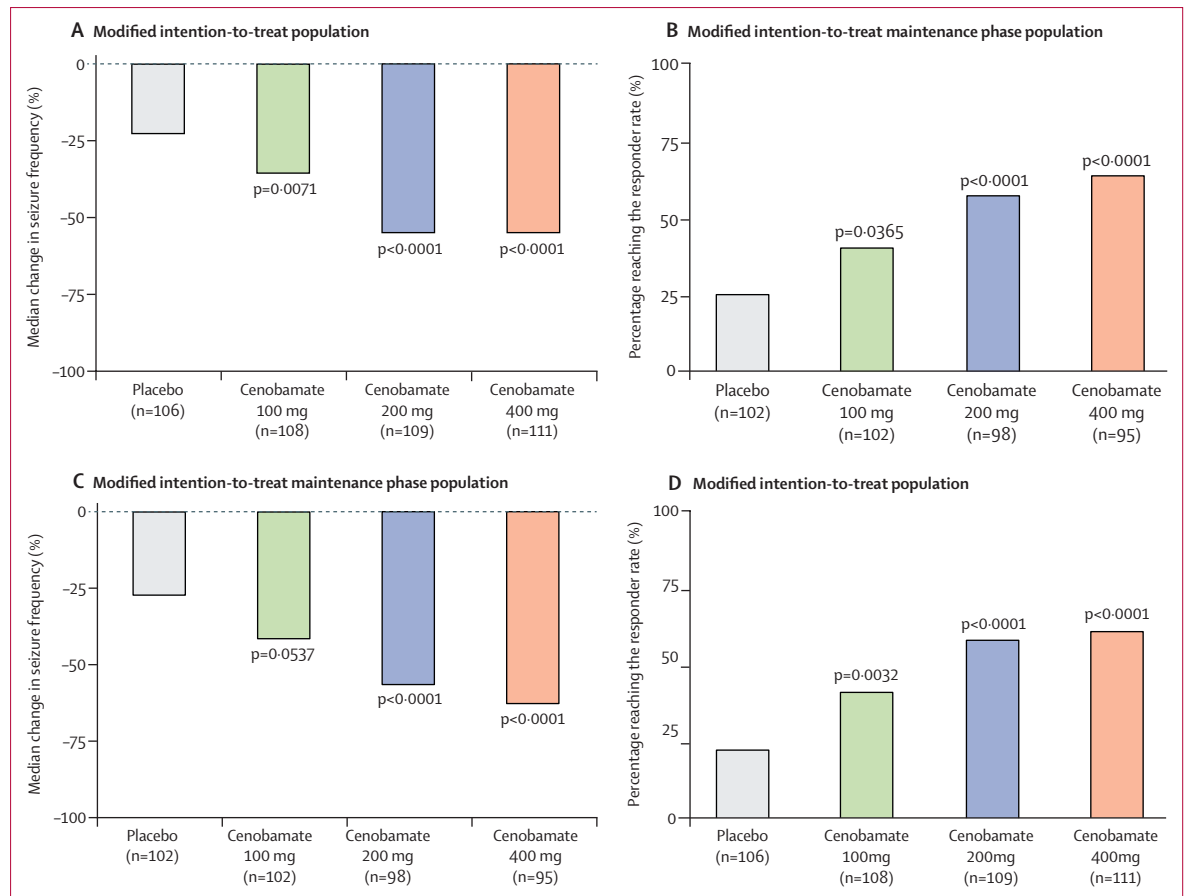


Figure 3: Primary and secondary efficacy analyses of median percentage reduction in 28-day focal seizure frequency (A, C) and primary and secondary efficacy analyses of 50% or more reduction in focal seizure frequency (B, D)

p values of the cenobamate treatment groups are those compared with the placebo group. Responder rates are the percentage of patients achieving a 50% or more reduction from baseline in focal seizure frequency.

had final responsibility for the decision to submit for publication.

Results

Between July 31, 2013, and June 22, 2015, a total of 533 patients were screened, of which 437 were randomly allocated to the treatment groups in this study at 107 epilepsy and general neurology centres in 16 countries (Australia [n=24], Bulgaria [n=37], Czech Republic [n=24], France [n=4], Germany [n=33], Hungary [n=15], Israel [n=12], Poland [n=48], Romania [n=4], Serbia [n=28], South Korea [n=31], Spain [n=27], Thailand [n=8], Ukraine [n=23], UK [n=8], and USA [n=111]; figure 2). Patients were randomly allocated to the treatment groups with cenobamate at 100 mg (n=108), 200 mg (n=110), or 400 mg (n=111), or the placebo group (n=108) and entered the titration phase (figure 2). The modified intention-to-treat population composed of 434 patients, the modified intention-to-treat maintenance phase population composed of 397, and the safety population composed of 437. 360 (82%) of 437 patients completed the study: 95 (88%) of 108 for the 100 mg dose group, 90 (82%) of

110 for the 200 mg dose group, 81 (73%) of 111 for the 400 mg dose group, and 94 (87%) of 108 for the placebo group. Adverse events were the primary reason for discontinuation in 12 (11%) patients in the 100 mg group, 15 (14%) in the 200 mg group, 23 (21%) in the 400 mg group, and five (5%) in the placebo group.

Baseline demographic and epilepsy-related characteristics were generally similar across groups (table 1). Of the 437 patients, 322 (74%) were taking two or three concomitant antiepileptic drugs. The most frequently used concomitant antiepileptic drugs (ie, $\geq 20\%$) were levetiracetam (186 [43%] of 437), lamotrigine (138 [32%]), and carbamazepine (122 [28%]; table 1).

In the modified intention-to-treat population, the median percentage change from baseline in focal seizure frequency per 28 days during the double-blind treatment was -24.0% (IQR -45 to -7.0%) for the placebo group, -35.5% (-62.5 to -15.0% ; $p=0.0071$) for the cenobamate 100 mg group, and -55.0% for both cenobamate 200 mg group (-73.0% to -23.0% ; $p<0.0001$) and 400 mg group (-85.0% to -28.0% ; $p<0.0001$; figure 3A; all p values vs placebo). During the 12-week maintenance phase (ie, the

modified intention-to-treat maintenance phase population), 25% (26 of 102 patients) in the placebo group had a 50% or more reduction in seizure frequency per 28 days (ie, responders) compared with 40% (41 of 102; odds ratio [OR] 1.97, 95% CI 1.08–3.56; $p=0.0365$) for the cenobamate 100 mg group, 56% (55 of 98; 3.74, 2.06–6.80; $p<0.0001$) for the cenobamate 200 mg group, and 64% (61 of 95; 5.24, 2.84–9.67; $p<0.0001$) for the cenobamate 400 mg group (figure 3B; all odds ratios and p values vs placebo). During the 12-week maintenance phase, the median percentage change in seizure frequency was -27.0% (IQR -51.0 to -8.0%) for the placebo group, -41.5% for the cenobamate 100 mg group (-65.0 to -11.0% ; $p=0.0537$), -56.5% for the cenobamate 200 mg group (-78.0 to -25.0% ; $p<0.0001$), and -63.0% for the cenobamate 400 mg group (-97.0 to -28.0% ; $p<0.0001$; figure 3C; all p values vs placebo). Results of a secondary responder analysis during the 18-week double-blind period (including the titration phase) were similar to those observed during the maintenance phase, showing an increasing percentage of responders with increasing dose (figure 3D; all p values vs placebo).

Seizure frequency reductions of 75% or more, 90% or more, and 100% during the 12-week maintenance phase (ie, the modified intention-to-treat maintenance phase population) were 10% ($n=10$), 3% ($n=3$), and 1% ($n=1$), respectively, for the placebo group; 17% ($n=17$; $p=0.2146$), 9% ($n=9$; $p=0.1340$), and 4% ($n=4$; $p=0.3688$) for the cenobamate 100 mg group; 31% ($n=30$; $p=0.0003$), 17% ($n=17$; $p=0.0007$), and 11% ($n=11$; $p=0.0022$) for the cenobamate 200 mg group; and 46% ($n=44$; $p<0.0001$), 28% ($n=27$; $p<0.0001$), and 21% ($n=20$; $p<0.0001$) for the cenobamate 400 mg group (figure 4; all p values vs placebo).

The median modal oral daily dose of cenobamate (ie, the most frequent dose received during the double-blind period) was 100 mg (IQR 100–100 mg) for the 100 mg dose group, 200 mg (200–200 mg) for the 200 mg dose group, and 300 mg (150–400 mg) for the 400 mg dose group. During the maintenance phase, in which patients received the maximum dose attained during titration, the median modal oral daily dose was 100 mg (100–100 mg) for the 100 mg dose group, 200 mg (200–200 mg) for the 200 mg dose group, and 400 mg (200–400) for the 400 mg dose group. Mean steady-state plasma concentrations at week 10 during the maintenance phase were $8.18 \mu\text{g/mL}$ (SD 4.68) for the 100 mg cenobamate group, $15.51 \mu\text{g/mL}$ (5.05) for the 200 mg group, and $23.46 \mu\text{g/mL}$ (10.07) for the 400 mg group. Similar steady-state plasma concentrations were observed at week 14. The non-linear increases in plasma concentration observed in the 400 mg group possibly occurred because some patients received doses less than 400 mg.

The percentage of patients who had at least one treatment-emergent adverse event during the double-blind treatment period was 70% (76 of 108) in the placebo group, 65% (70 of 108) in the 100 mg cenobamate group, 76% (84 of 110) in the 200 mg group, and 90% (100 of 111)

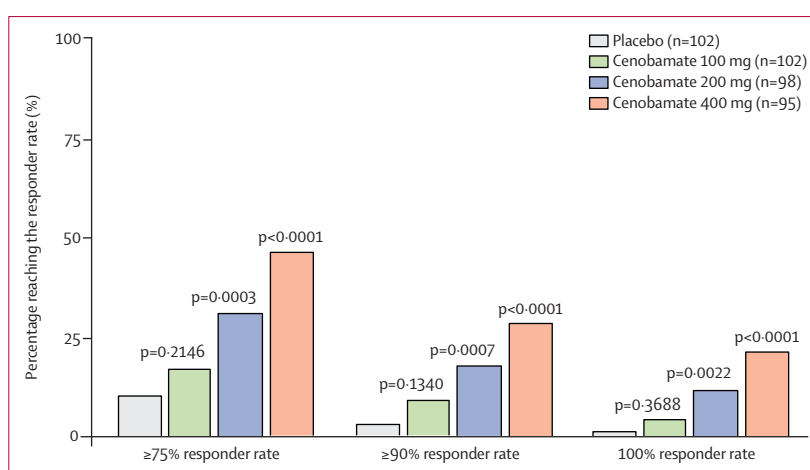


Figure 4: 75% or more, 90% or more, and 100% seizure responder rates during the maintenance phase (ie, the modified intention-to-treat maintenance phase population)

p values of the cenobamate treatment groups are those compared with the placebo group. Responder rates are the percentage of patients achieving a percentage reduction from baseline in focal seizure frequency.

	Placebo (n=108)	Cenobamate		
		100 mg (n=108)	200 mg (n=110)	400 mg (n=111)
Any treatment-emergent adverse events	76 (70%)	70 (65%)	84 (76%)	100 (90%)
Treatment-related adverse events	46 (43%)	62 (57%)	72 (65%)	92 (83%)
Severe treatment-emergent adverse events	8 (7%)	11 (10%)	11 (10%)	18 (16%)
Serious treatment-emergent adverse events	6 (6%)	10 (9%)	4 (4%)	8 (7%)
Adverse events $\geq 5\%$				
Somnolence	9 (8%)	20 (19%)	23 (21%)	41 (37%)
Dizziness	15 (14%)	19 (18%)	22 (20%)	37 (33%)
Headache	6 (6%)	11 (10%)	12 (11%)	12 (11%)
Balance disorder	0	3 (3%)	2 (2%)	10 (9%)
Nystagmus	1 (<1%)	3 (3%)	4 (4%)	7 (6%)
Ataxia	1 (<1%)	2 (2%)	4 (4%)	7 (6%)
Dysarthria	0	2 (2%)	3 (3%)	7 (6%)
Fatigue	9 (8%)	13 (12%)	19 (17%)	27 (24%)
Gait disturbance	3 (3%)	1 (<1%)	6 (6%)	9 (8%)
Diplopia	2 (2%)	8 (7%)	11 (10%)	17 (15%)
Constipation	1 (<1%)	2 (2%)	3 (3%)	10 (9%)
Nausea	1 (<1%)	7 (7%)	1 (<1%)	10 (9%)
Vomiting	0	2 (2%)	3 (3%)	6 (5%)
Fall	6 (6%)	2 (2%)	4 (4%)	4 (4%)
Upper respiratory tract infection	6 (6%)	3 (3%)	4 (4%)	3 (3%)
Back pain	3 (3%)	4 (4%)	1 (<1%)	6 (5%)
Vertigo	3 (3%)	1 (<1%)	3 (3%)	6 (5%)
Decreased appetite	1 (<1%)	3 (3%)	1 (<1%)	6 (5%)

Data are n (%).

Table 2: Summary of adverse events in the safety population

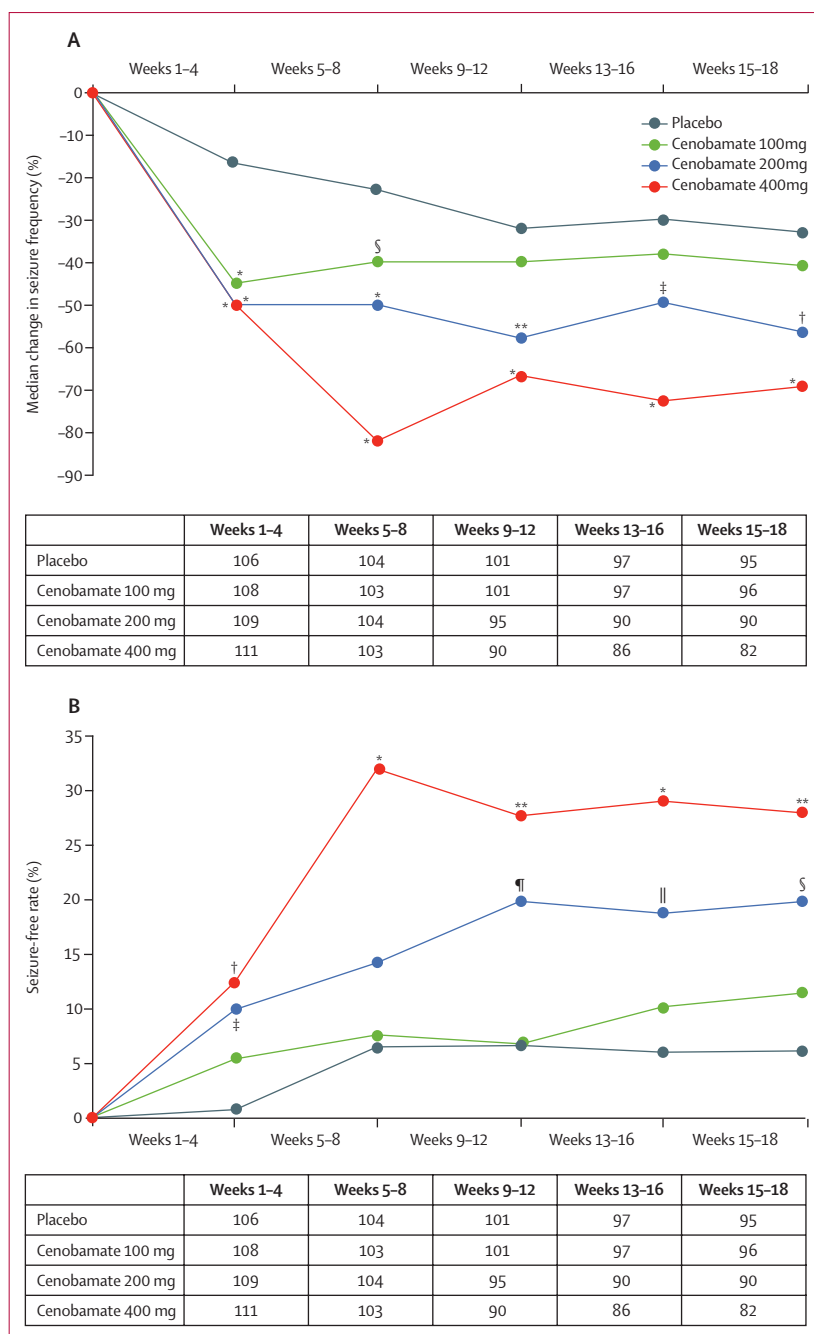


Figure 5: Post-hoc analyses of the median percentage reduction in seizure frequency (A) and seizure-free rate (B) over time using 4-week intervals in the modified intention-to-treat population. Weeks 15-18 overlap in order to make the interval 4 weeks in duration. Panel A: * $p < 0.0001$, ** $p = 0.0001$, † $p = 0.0004$, ‡ $p = 0.0011$, § $p = 0.0461$, all vs placebo. Panel B: * $p < 0.0001$, ** $p = 0.0002$, † $p = 0.0007$, ‡ $p = 0.0051$, § $p = 0.0078$, ¶ $p = 0.0105$, || $p = 0.0129$, all vs placebo. All datapoints without a symbol were not significant.

in the 400 mg group (table 2). The most frequently reported ($\geq 10\%$) treatment-emergent adverse events with cenobamate were somnolence, dizziness, headache, fatigue, and diplopia, in which the incidences increased with dose (table 2). Dizziness was the most frequently reported treatment-emergent adverse event in the placebo

group. Most treatment-emergent adverse events were mild or moderate in severity (68 [63%] of 108 patients in the placebo group, 59 [55%] of 108 in the 100 mg group, 73 [66%] of 110 in the 200 mg group, and 82 [74%] of 111 in the 400 mg group). There were 37 serious treatment-emergent adverse events reported in 28 patients, including six (6%) of 108 patients in the placebo group, ten (9%) of 108 in the 100 mg group, four (4%) of 110 in the 200 mg group, and eight (7%) of 111 in the 400 mg group. There were no deaths. Serious treatment-emergent adverse events occurring in more than one patient in any treatment group were seizures ($n=2$; cenobamate 100 mg), ataxia ($n=2$; cenobamate 400 mg), dizziness ($n=2$; cenobamate 400 mg), nystagmus ($n=2$; cenobamate 400 mg), and suicidal ideation ($n=2$; cenobamate 100 mg). One suicide attempt occurred, in a 35-year-old female patient in the cenobamate 100 mg group; this patient had an ongoing mood disorder and was receiving concomitant escitalopram (20 mg/day). Her concomitant antiepileptic drugs included lamotrigine, levetiracetam, and clobazam. A total of 53 patients had treatment-emergent adverse events that led to treatment discontinuation, including five (5%) of 108 patients treated with placebo and 11 (10%) of 108 treated with cenobamate 100 mg/day, 15 (14%) of 110 with 200 mg/day, and 22 (20%) of 111 with 400 mg/day. Treatment-emergent adverse events that led to discontinuation in more than one patient in any group were ataxia ($n=3$, cenobamate 200 mg; $n=4$, cenobamate 400 mg), dizziness ($n=4$, cenobamate 400 mg), somnolence ($n=2$, cenobamate 200 mg; $n=3$, cenobamate 400 mg), nystagmus ($n=3$, cenobamate 400 mg), and vertigo ($n=3$, cenobamate 400 mg).

Three hypersensitivity reactions, characterised as rash with involvement of at least one other body system, were reported in three patients treated with cenobamate. Two of the reactions were non-serious cases of rash of moderate severity that occurred in two patients randomly assigned to receive 200 mg/day of cenobamate. One case (pruritic rash) occurred on day 10 of treatment during the initial faster 100 mg/week titration. This event was accompanied by pyrexia. In the other case, a rash and facial swelling occurred on day 57 in a patient who underwent the amended titration protocol. In both cases, treatment was discontinued and the rash resolved. The third hypersensitivity reaction was a serious case of drug reaction with eosinophilia and systemic symptoms (DRESS) that occurred starting on day 24 of treatment in a patient randomly assigned to receive 200 mg/day of cenobamate during the faster initial titration protocol. Treatment was discontinued and the patient was treated with corticosteroids and recovered within 2 months.

No clinically meaningful results were observed in changes from baseline in haematology, clinical chemistry, laboratory values, ECG studies, vital signs, or physical or neurological examinations.

Post-hoc analyses of focal seizure subtypes found evidence of benefit with cenobamate across all evaluated

seizure types (appendix pp 1, 2). The median percentage change in focal aware motor seizures per 28 days during the maintenance phase was 11.0% (IQR -21.0 to 44.0%) for the placebo group and -49.0% (-73.0 to -11.0%; $p=0.0012$) for the cenobamate 100 mg group, -62.0% (-74.5 to -18.5%; $p=0.0010$) for the 200 mg group, and -69.0% (-89.0 to -34.5%; $p=0.0001$) for the 400 mg group (appendix p 1). Focal seizures with impaired awareness and focal to bilateral tonic-clonic seizures were substantially reduced in the 200 mg and 400 mg cenobamate dose groups versus placebo (appendix pp 1, 2). Post-hoc analyses of responder rates ($\geq 75\%$, $\geq 90\%$, and 100% reduction) by seizure types provided evidence of a similar positive benefit with cenobamate compared with placebo in each dose group (appendix pp 3, 4). An alternative method of analysing seizure freedom was applied post hoc using patients who were seizure-free during the maintenance phase and who completed the study (excluding dropouts) as the numerator and the modified intention-to-treat population as the denominator.¹⁷ In this analysis, less than 1% (one of 106) of patients in the placebo group achieved seizure freedom during the maintenance phase compared with 3% (3 of 108) of patients in the cenobamate 100 mg group, 9% (ten of 109) in the 200 mg group, and 14% (16 of 111) in the 400 mg group. An exploratory post-hoc analysis using the modified intention-to-treat maintenance phase population determined that the number needed to treat to achieve seizure-freedom was 10 in the cenobamate 200 mg group and five in the 400 mg group.

During the first 4 weeks of the double-blind treatment, the median seizure frequency reduction was 17.0% (IQR 8.0–47.0%) in the placebo group versus 45% (11.5–67.0%) for the cenobamate 100 mg group and 50.0% for both the 200 mg (IQR 17.0–75.0%) and 400 mg (15.0–78.0%) groups (figure 5A). Sustained decreases in median seizure frequency were noted at each additional 4-week interval in the 200 mg and 400 mg cenobamate dose groups. In addition, high rates of 100% seizure reduction occurred in the 200 mg and 400 mg cenobamate dose groups within each 4-week interval starting at weeks 5–8 (figure 5B).

Discussion

This multicentre, randomised study of patients with uncontrolled focal seizures showed that adjunctive therapy with cenobamate at doses of 100 mg/day, 200 mg/day, and 400 mg/day led to marked reductions in focal seizure frequency from baseline during the 18-week double-blind treatment period, with the greatest reduction occurring in the 200 mg and 400 mg dose groups. Because patients in the 400 mg group completed titration at the beginning of the maintenance phase, the analysis of efficacy during the maintenance phase should better reflect the potential efficacy in this group. This analysis showed a higher median percentage of seizure frequency reduction in the 400 mg group (63.0%) compared with the 100 mg (41.5%)

and 200 mg groups (56.5%), suggesting a dose-related improvement. A similar dose-effect was seen for the 50% or more seizure responder outcome, with responder rates ranging from 40% to 64%. Post-hoc analyses also provided evidence that seizure frequencies decreased relatively early during cenobamate titration; seizure reductions of 45–50% were evident across dose groups during the first 4 weeks of 50–100 mg/week titration.

Measured only over a 12-week maintenance phase, significantly more patients treated with 200 mg and 400 mg of cenobamate achieved seizure freedom (100% reduction) than those given placebo. This effect also appeared to be dose-related (although not significantly): in the 100 mg, 200 mg, and 400 mg groups, respectively, 4%, 11%, and 21% of patients were seizure-free during the maintenance phase (ie, the modified intention-to-treat maintenance phase population), compared with 1.0% of placebo-treated patients. This benefit was supported by a conservative analysis of seizure-free rates during the maintenance phase among patients completing the study (preventing dropouts from counting as seizure-free) and using the modified intention-to-treat population as the denominator. This analysis also noted high seizure-free rates for patients treated with 200 mg (9.2%) and 400 mg (14.4%). Regardless of the analysis method used, it remains difficult to interpret seizure freedom in clinical trials given the constraints of the study designs (ie, inability to adjust concomitant antiepileptic drugs), which do not reflect real-life practice. Nonetheless, seizure freedom is of great clinical significance to patient quality of life^{5,18} and the rates reported in this study are notable relative to all other pivotal studies of antiepileptic drug treatment in uncontrolled focal seizures over the past 25 years.^{13–16,19–28} The ongoing open-label study will help address how long patients maintain seizure freedom or clinically significant seizure control with cenobamate. Considering the high rates of seizure freedom observed in this study, cenobamate's mechanisms of action and its potential effects on concomitant antiepileptic drugs also warrant further investigation.

Reductions in focal aware with motor component, focal with impaired awareness, or focal to bilateral tonic-clonic seizures were also noted in this study, with the highest reduction occurring in focal to bilateral tonic-clonic seizures. Additional benefit was observed with the 400 mg dose group in reducing focal aware motor and focal impaired awareness seizures.

The most commonly reported treatment-emergent adverse events associated with cenobamate were CNS-related (somnolence, dizziness, fatigue), similar to those with other antiepileptic drugs, and the incidence increased with the dose. The majority of events, however, were mild or moderate in severity. Serious adverse event incidences ranged from 4% in the 200 mg group to 9% in the 100 mg group, compared with 6% in the placebo group. Discontinuations due to adverse events also increased in a dose-related manner, with 21% in the 400 mg

group discontinuing treatment. The rapid titration of 100 mg/week from 200 mg to 400 mg might have contributed, at least in part, to the higher rates of treatment-emergent adverse events in the 400 mg group. Additionally, because this study was an adjunctive, placebo-controlled treatment trial, patients were not able to adjust the dose of concomitant antiepileptic drug treatments when titrating to higher doses, which might have also contributed to treatment-emergent adverse events (65% of patients in the 200 mg and 78% of those in the 400 mg group were taking two or three concomitant antiepileptic drugs). Three cases of readily identifiable hypersensitivity adverse events occurred in the 200 mg dose group, including one serious event of DRESS. Two of three of these events, including the case of DRESS, occurred in patients who were on the original faster titration schedule. Cutaneous, idiosyncratic adverse events have been reported with antiepileptic drugs, including DRESS and Stevens–Johnson syndrome, which can be severe and life-threatening.²⁹ Slowing the titration rate has been shown to reduce the risk of skin hypersensitivity reactions with other antiepileptic drugs.³⁰ A large, ongoing phase 3 study (NCT02535091) of cenobamate is investigating the use of a lower starting dose (12·5 mg) and slower titration rate, with dose increases every 2 weeks (25 mg, 50 mg, 100 mg, 150 mg, and 200 mg).

Limitations of this study, as with other controlled phase 2 and 3 studies of adjunctive antiepileptic drugs, include the short study duration and the potential effect of concomitant medications. The use of a placebo group, although still favoured in adjunctive antiepileptic drug clinical studies for assessment of safety, precludes longer treatment durations for assessment of seizure freedom, because of ethical considerations. The ongoing open-label extension phase of this study will provide additional insight into the long-term safety and efficacy profile of adjunctive cenobamate. Pharmacokinetic analyses will also help to inform the effects of cenobamate on concomitant antiepileptic drugs. Other limitations include the use of self-recorded seizure frequency and seizure type, which might be considered as a potential source of bias. This study design is, however, the current standard for antiepileptic drug studies despite the limitations.

In summary, this study shows that adjunctive therapy with cenobamate at 100 mg/day, 200 mg/day, and 400 mg/day significantly improved seizure control in patients with focal epilepsy who were not adequately treated with one to three antiepileptic drugs. Efficacy with cenobamate occurred in a dose-related manner and treatment was generally well tolerated. High rates of seizure freedom were observed with doses of 200 mg and 400 mg. These data suggest that cenobamate might be a safe and effective treatment option in patients with uncontrolled focal (partial)-onset seizures.

Contributors

MK conceived and designed the study. GLK, PK, CB, SKL, IM, MM, and BJS acquired data, and all authors had access to and interpreted the raw

data. All authors reviewed and revised drafts of the manuscript and approved the final version.

Declaration of interests

GLK is a consultant or advisor for Eisai, Otsuka, and Shire; and has received research support from Biogen, SK life science, and UCB Pharma. PK is a consultant for AbbVie, Alliance, Eisai, Engage, and UCB Pharma; has participated in advisory boards of Aquestive and SK life science; has been a speaker for Aquestive, Eisai, Sunovion, and UCB Pharma; and has received research support from Eisai and Lundbeck. CB is or was a consultant or advisor for UCB Pharma, Eisai, Novartis, Desitin, Idorsia, and GW Pharmaceuticals; and has been a speaker for UCB Pharma, Eisai, Upsher-Smith, and SK life science. BJS was or is a consultant or advisor for B Braun Melsungen, Desitin, Eisai, GW Pharmaceuticals, Idorsia, and UCB Pharma; and has received research support from Bial, Desitin, Eisai, GW Pharmaceuticals, Hexal, Novartis, SK life science, and UCB Pharma. MK is an employee of SK life science. SKL, IM, and MM declare no competing interests.

Data sharing

The data from this study are available by request from the authors or SK life science (Paramus, NJ, USA), the company sponsoring the clinical development of cenobamate for the treatment of focal epilepsy. At the time of the request, the format and scope of the data to be disseminated will be determined by the authors and SK life science.

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References

- 1 Fountain NB, Van Ness PC, Bennett A, et al. Quality improvement in neurology: Epilepsy Update Quality Measurement Set. *Neurology* 2015; **84**: 1483–87.
- 2 Chen Z, Brodie MJ, Liew D, Kwan P. Treatment outcomes in patients with newly diagnosed epilepsy treated with established and new antiepileptic drugs: a 30-year longitudinal cohort study. *JAMA Neurol* 2018; **75**: 279–86.
- 3 Tian N, Boring M, Kobay R, Zack MM, Croft JB. Active epilepsy and seizure control in adults—United States, 2013 and 2015. *MMWR Morb Mortal Wkly Rep* 2018; **67**: 437–42.
- 4 Leppik I, De Rue K, Edrich P, Perucca E. Measurement of seizure freedom in adjunctive therapy studies in refractory partial epilepsy: the levetiracetam experience. *Epileptic Disord* 2006; **8**: 118–30.
- 5 Josephson CB, Patten SB, Bulloch A, et al. The impact of seizures on epilepsy outcomes: a national, community-based survey. *Epilepsia* 2017; **58**: 764–71.
- 6 Thurman DJ, Logroscino G, Beghi E, et al. The burden of premature mortality of epilepsy in high-income countries: a systematic review from the Mortality Task Force of the International League Against Epilepsy. *Epilepsia* 2017; **58**: 17–26.
- 7 Hesdorffer DC, Tomson T, Benn E, et al. Combined analysis of risk factors for SUDEP. *Epilepsia* 2011; **52**: 1150–59.
- 8 Bialer M, Johannessen SI, Levy RH, Perucca E, Tomson T, White HS. Progress report on new antiepileptic drugs: a summary of the Eleventh Eilat Conference (EILAT XI). *Epilepsy Res* 2013; **103**: 2–30.
- 9 Nakamura M, Cho JH, Shin H, Jang IS. Neuropharmacology and analgesia effects of cenobamate (YKP3089), a newly developed anti-epileptic drug, on voltage-gated sodium channels in rat hippocampal CA3 neurons. *Eur J Pharmacol* 2019; **855**: 175–82.
- 10 International Council for Harmonisation. Integrated Addendum to ICH E6(R1): guideline for good clinical practice E6(R2). Nov 9, 2016. https://www.ich.org/fileadmin/Public_Web_Site/ICH_Products/Guidelines/Efficacy/E6/E6_R2_Step_4_2016_1109.pdf (accessed Feb 25, 2019).
- 11 International League Against Epilepsy. Proposal for revised clinical and electroencephalographic classification of epileptic seizures. From the Commission on Classification and Terminology of the International League Against Epilepsy. *Epilepsia* 1981; **22**: 489–501.
- 12 Berg AT, Berkovic SF, Brodie MJ, et al. Revised terminology and concepts for organization of seizures and epilepsies: report of the ILAE Commission on Classification and Terminology, 2005–2009. *Epilepsia* 2010; **51**: 676–85.

- 13 Gil-Nagel A, Zaccara G, Baldinetti F, Leon T. Add-on treatment with pregabalin for partial seizures with or without generalisation: pooled data analysis of four randomised placebo-controlled trials. *Seizure* 2009; **18**: 184–92.
- 14 French JA, Krauss GL, Biton V, et al. Adjunctive perampanel for refractory partial-onset seizures: randomized phase III study 304. *Neurology* 2012; **79**: 589–96.
- 15 Krauss GL, Serratosa JM, Villanueva V, et al. Randomized phase III study 306: adjunctive perampanel for refractory partial-onset seizures. *Neurology* 2012; **78**: 1408–15.
- 16 French JA, Krauss GL, Steinhoff BJ, et al. Evaluation of adjunctive perampanel in patients with refractory partial-onset seizures: results of randomized global phase III study 305. *Epilepsia* 2013; **54**: 117–25.
- 17 Gazzola DM, Balcer LJ, French JA. Seizure-free outcome in randomized add-on trials of the new antiepileptic drugs. *Epilepsia* 2007; **48**: 1303–07.
- 18 Birbeck GL, Hays RD, Cui X, Vickrey BG. Seizure reduction and quality of life improvements in people with epilepsy. *Epilepsia* 2002; **43**: 535–38.
- 19 Costa J, Fareleira F, Ascencao R, Borges M, Sampaio C, Vaz-Carneiro A. Clinical comparability of the new antiepileptic drugs in refractory partial epilepsy: a systematic review and meta-analysis. *Epilepsia* 2011; **52**: 1280–91.
- 20 Ryvlin P, Werhahn KJ, Blaszczyk B, Johnson ME, Lu S. Adjunctive brivaracetam in adults with uncontrolled focal epilepsy: results from a double-blind, randomized, placebo-controlled trial. *Epilepsia* 2014; **55**: 47–56.
- 21 Biton V, Berkovic SF, Abou-Khalil B, Sperling MR, Johnson ME, Lu S. Brivaracetam as adjunctive treatment for uncontrolled partial epilepsy in adults: a phase III randomized, double-blind, placebo-controlled trial. *Epilepsia* 2014; **55**: 57–66.
- 22 Klein P, Schiemann J, Sperling MR, et al. A randomized, double-blind, placebo-controlled, multicenter, parallel-group study to evaluate the efficacy and safety of adjunctive brivaracetam in adult patients with uncontrolled partial-onset seizures. *Epilepsia* 2015; **56**: 1890–98.
- 23 Biton V, Rogin JB, Krauss G, et al. Adjunctive eslicarbazepine acetate: a pooled analysis of three phase III trials. *Epilepsy Behav* 2017; **72**: 127–34.
- 24 Chung SS, Fakhoury TA, Hogan RE, et al. Once-daily USL255 as adjunctive treatment of partial-onset seizures: randomized phase III study. *Epilepsia* 2014; **55**: 1077–87.
- 25 French JA, Baroldi P, Brittain ST, Johnson JK. Efficacy and safety of extended-release oxcarbazepine (Oxtellar XR) as adjunctive therapy in patients with refractory partial-onset seizures: a randomized controlled trial. *Acta Neurol Scand* 2014; **129**: 143–53.
- 26 Peeters K, Adriaenssen I, Wapenaar R, Neto W, Pledger G. A pooled analysis of adjunctive topiramate in refractory partial epilepsy. *Acta Neurol Scand* 2003; **108**: 9–15.
- 27 Steinhoff BJ, Ben-Menachem E, Ryvlin P, et al. Efficacy and safety of adjunctive perampanel for the treatment of refractory partial seizures: a pooled analysis of three phase III studies. *Epilepsia* 2013; **54**: 1481–89.
- 28 Ben-Menachem E, Mameniski R, Quarato PP, et al. Efficacy and safety of brivaracetam for partial-onset seizures in 3 pooled clinical studies. *Neurology* 2016; **87**: 314–23.
- 29 Zaccara G, Franciotta D, Perucca E. Idiosyncratic adverse reactions to antiepileptic drugs. *Epilepsia* 2007; **48**: 1223–44.
- 30 Hirsch LJ, Weintraub DB, Buchsbaum R, et al. Predictors of lamotrigine-associated rash. *Epilepsia* 2006; **47**: 318–22.